MIL-L-403D
19 August 1969
SUPERSEDING
MIL-L-403C
9 April 1965

MILITARY SPECIFICATION

LOADERS, BUCKET-TYPE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers gasoline- and diesel-engine-driven, crawler-mounted, bucket-type loaders.

1.2 Classification. Loaders shall be of the following classes, as specified:

Class 1 - Diesel-engine-driven.
Class 2 - Gasoline-engine-driven.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

O-A-548	- Antifreeze, Ethylene Glycol, Inhibited.
VV-F-800	- Fuel Oil, Diesel.

Military

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.

FSC 3805

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MIL-I-631	- Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine (Heavy Duty).
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-T-3351	- Tractor, Full-Tracked, Low-Speed; Tractor, Wheeled, Agriculture; and Tractor, Wheeled, Industrial: And Their Attachments, Packaging of.
MIL-L-6085	- Lubricating Oil: Instrument, Aircraft, Low Volatility.
MIL-L-10295	- Lubricating Oil, Internal-Combustion Engine, Sub-Zero.
MIL-L-10324	- Lubricating Oil, Gear, Sub-Zero.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-E-11275	- Engines, Gasoline, Industrial Type, General Specification for.
MIL-E-11276	- Engines, Diesel: Industrial, Medium- and High-Speed, General Specification for.
MIL-A-11755	- Antifreeze, Arctic-Type.
MIL-C-13486	- Cable, Special Purpose, Electrical: Low-Tension, Heavy-Duty Single-Conductor and Multiconductor.
MIL-I-15126	- Insulation Tape, Electrical, Pressure Sensitive Adhesive and Pressure Sensitive Thermosetting Adhesive.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-L-45199	- Lubricating Oil, Internal Combustion Engine (High Output Diesel).
MIL-T-46786	- Tanks, Fuel, Engine, General Requirements for.

STANDARDS

Military

MIL-STD-130	- Identification Marking of U. S. Military Property.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.

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MIL-STD-646	- Electrical Circuit (Wire Marking) Identification for Tactical Military Vehicles.
MIL-STD-1400	- Engines, Gasoline or Diesel, Methods of Test.
MS35000	- Battery, Storage, Lead-Acid, Waterproof.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY

Standard Qualification Procedure.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, N. Y. 10001.)

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3. REQUIREMENTS

3.1 Description. Loaders shall be self-feeding, self-propelled, crawler-mounted, bucket-type and shall pick up and convey all types of graded aggregate and other types of loose bulk material from the ground or a stock pile and discharge it through a chute or spout. Each loader shall discharge free-flowing aggregate at a rate of not less than 3 cubic yards per minute.

3.2 Preproduction model. The supplier shall furnish a loader for examination and test within the time frame specified to prove, prior to starting production, that his production methods and choice of design detail will produce loaders that comply with the requirements of this specification (see 6.3). Examination and tests shall be those specified herein.

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the supplier and shall be subject to all the provisions of this specification. Materials and special treatments shall be utilized to prevent corrosion and to resist harmful effects of dust, fresh and salt water, and fungi.

3.4 Screw threads. Screw threads shall be in accordance with Handbook H28.

3.5 Environmental requirements.

3.5.1 Temperature. Without preheating, the engine shall start within 5 minutes and the loader shall operate under full-load conditions within 15 minutes after engine start in any ambient temperature from minus 25° F. to plus 120° F. All low temperature starting and operating controls shall be such that they can be manipulated by operators wearing arctic clothing.

3.5.2 Precipitation. The loader shall start and operate when exposed to heavy rain and high winds (30 mph) during both operation and extended outside storage. The loader shall start within 2 minutes and operate during and after exposure to rain and wind when tested as specified in 4.5.2.8. The loader shall operate as follows:

(a) Without malfunction of any electrical component, gage, or instrument.
(b) Without leakage of water into the exhaust system or into the air-intake system in excess of 8 percent of the fluid capacity of the sediment chamber.
(c) Without leakage of water into the engine crankcase sump, transmission, steering clutch and brake compartments, bevel gear and final drive compartment, hydraulic reservoir, or fuel tank.

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3.6 Safety. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are of such nature or so located as to be a hazard to operating personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions.

3.7 Maintenance. All assemblies and installed attachments shall be accessible for servicing, repair, and replacement. Sizes and types of connectors, plugs, and valves shall be held to a minimum.

3.7.1 Detachable fittings. Disconnect plugs and receptacles, multiple-line connectors in the electrical system, and detachable and attachable-type fittings in the pneumatic system shall be utilized, where applicable, to permit component removal and replacement.

3.7.2 Filling, draining, and checking provisions. Transmission housings shall be equipped with dipsticks to determine the fluid level. Other enclosures such as gear cases and reservoirs which contain a reservoir of lubricant or hydraulic oil shall be equipped with dipsticks, or with check plugs of not less than 1/2-inch pipe size, to determine the fluid level. Each enclosure shall be equipped with a means for filling and for draining. Except when an integral tube is provided, the drain outlet shall be fitted with a magnetic drain plug having either tapered pipe threads or shoulder and gasket; or in lieu of the magnetic drain plug, a separate magnetic device shall be fastened inside the enclosure at or near the bottom and shall be removable for cleaning without removal of other parts or components. Each drain plug or valve shall be located so that removal of the plug or opening of the valve will result in complete drainage of the fluid from the enclosure. When the loader is in a level position, drainage of fluids shall be to the ground without draining on or over any part of the loader, except the portion of the track which is in contact with the ground. Integral tubes, hose, or troughs may be used to convey the fluids from the drain plugs or valves. Accessibility to the filling means, the fluid-level checking device, and the drain plug or valve shall be provided without the removal or adjustment of accessories and parts, except plates equipped with hand-operated, quick-disconnect fasteners, or cap screws in applications where quick-disconnect fasteners are subject to damage. The drains for the engine-cooling system shall provide complete drainage of the system. The radiator drain shall be such that coolant will not drain onto any part of the loader. Extension hoses from the drain valves are permitted.

3.8 Main transmission. The main transmission shall be of the heavy-duty type and shall connect to the engine through a friction clutch or torque converter permitting not less than two forward speeds and one reverse speed.

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3.8.1 Travel speeds. The forward travel speed shall range from not more than 25 feet per minute (fpm) to not less than 175 fpm, and reverse travel speed shall be approximately 60 fpm at engine governed speed.

3.8.2 Feeder transmission and clutch. The crowd speed of the crawlers shall be controlled by a feeder transmission which will provide a total of not less than 12 operating speeds by the use of interchangeable gears. Two speeds shall be obtained with each set of gears. The crowd speeds shall range from not more than 3 fpm to not less than 30 fpm at engine governed speed.

3.9 Crawlers. The crawler treads shall be not less than 13 inches wide and each shall have not less than 7 feet of ground contact. When specified, the crawlers shall be provided with bolt holes to permit attachment of grousers. The crawler drive shall permit either crawler to be braked, engaged, or driven through an independent friction clutch. The crawlers shall be constructed so that the horizontal position of the frame will not be changed when the loader is operating on an irregular grade. The loader shall turn through 360 degrees in a horizontal radius equal to the length of the ground contact of the crawlers.

3.10 Carrier frame. The carrier frame shall be of welded or riveted structural steel construction forming a base to support the loader parts. Bumper blocks shall be secured to the carrier frame to prevent trucks from damaging the loader.

3.11 Operator's platform. The operator's platform shall be constructed of steel and shall have a nonslip surface and a protective steel canopy. The platform shall be located so that the operator will have an optimum view of all operations and access to all controls. For access to the platform, grab bars and nonslip-surface steps shall be provided.

3.12 Electrical system. The loader shall be equipped with a 24-volt electrical system.

3.12.1 Wiring. Electrical cable shall conform to MIL-C-13486, type I, class A. Grommets shall be used to prevent chafing where cable or harnesses pass through any member. Cushioned support straps shall be used to support cable and harnesses, and no unsupported length shall exceed 36 inches. Wiring shall be number traced in accordance with MIL-STD-646. Terminal-type connectors with insulator sleeves shall be used on the floodlight and headlight connections. Cable splices shall be waterproof. Cable shall be bound together with one-half overlapping turns of 0.010-inch-thick, 1-inch-maximum-width, black, electrical insulating tape conforming to MIL-I-631, type F, form Ts, grade a, class I, or MIL-I-15126, type XX, as applicable.

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3.12.2 Lighting.

3.12.2.1 Sealed-beam units. Sealed-beam units for floodlights shall conform to trade number 4578.

3.12.2.2 Light housings. Floodlight housings and lamp housing, except for mounting brackets, shall be furnished. Mounting brackets shall be of steel not less than 3/16 inch thick and shall provide floodlight adjustments as follows:

(a) Front floodlight. Not less than 10 degrees above and below the horizontal in the vertical plane and not less than 5 degrees to the right and left in the horizontal plane.

(b) Rear floodlights. Not less than 30 degrees above and below the horizontal in the vertical plane and not less than 180 degrees to the right and to the left in the horizontal plane.

3.12.2.3 Light-mounting location. Lights shall be mounted and adjusted as follows.

3.12.2.3.1 Rear floodlights. Two floodlights mounted on the chassis, one on the left and one on the right, shall be provided. The higher intensity light pattern shall be directed approximately parallel to the longitudinal axis of the loader to illuminate the loading space of both sides of the loader when its center is aimed at the material to be loaded.

3.12.2.3.2 Front floodlight. One floodlight, mounted on the upper front corner of the chassis frame, shall be provided. The floodlight shall be adjusted for distant viewing to the front of the loader. The housing shall be guarded.

3.12.2.3.3 Instrument panel light, or lights. An instrument panel light, or lights, shall be provided to illuminate all gages, instruments, and switches mounted on the panel.

3.12.2.4 Switches and circuit breakers. Lights shall be controlled from toggle switches. The circuits shall be provided with circuit breakers. All switches shall be mounted on the instrument panel. One switch and associated electrical circuit shall be provided for the independent control of each of the following light combinations:

(a) Forward floodlight.
(b) Rear floodlights.
(c) Instrument panel light, or lights.

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3.12.2.5 Trouble lamp. A trouble lamp complete with wire, plug, and matching weather protected socket shall be furnished and mounted on the instrument panel. The length of the wire shall be sufficient to allow the lamp to illuminate any part of the loader without being taut.

3.13 Boom assembly.

3.13.1 Boom. The boom shall be not less than 17 feet long from center-to-center of sprockets and constructed of trussed steel plate rigidly braced and welded or riveted. The boom shall be equipped with an enclosed steel chute to return spillage at the discharge end to the feeder mechanism. The boom shall be mounted to permit pivoting from the loading position to a horizontal position. In the horizontal position, the boom shall be not more than 12 feet in height. Power for raising and lowering the boom shall be controlled hydraulically or mechanically through a friction clutch. A manual means shall be provided to adjust accurately to grade requirements.

3.13.2 Conveyor. The conveyor shall consist of welded-steel buckets attached to the links of a roller chain and shall be provided with a stop to prevent roll back of the conveyor. A chain and sprocket drive shall provide a bucket-line speed of not less than 160 fpm. The bucket lines shall be engaged by a jaw or friction clutch, and provision shall be made for maintaining tension in the drive and bucket line.

3.13.3 Feeding system. A positive-action feeding system, including spirals or propellers, a scraper, and provision for automatic or manual resetting overload protection, shall be provided. When the overload-protection device is in a location not within reach of the operator in his normal operating position, it shall be of the automatic-resetting type.

3.13.3.1 Spirals or propellers. Feeder spirals shall be of steel construction hard-surface wearing blades. Propellers shall be of austenitic manganese steel provided with reversible, replaceable edges. Spirals or propellers shall be chain-driven.

3.13.3.2 Scraper. The loader shall be equipped with a full-width, three-piece floating scraper that will automatically conform to the transverse and longitudinal grade contour of the road without the necessity of manual control. Provision shall be made for bolting outrigger shoes to the outer side of the scraper (see 3.15.4). Connecting chains or solid adjustable bars shall be furnished for holding the scraper to the boom during travel.

3.14 Discharge spout. The discharge spout shall have a total swing of 180 degrees (90 degrees to either side of the longitudinal axis) and a discharge height of not less than 10 feet above the ground. The discharge spout shall be operated from the operator's platform.

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3.15 Accessories. When specified, the loader shall be equipped with any or all of the following accessories.

3.15.1 Adjustable discharge chute. The discharge chute shall direct the flow of material. The chute shall be pivoted or hinged and shall be adjustable in the vertical plane.

3.15.2 Housing for return bucket line. The housing shall be heavy canvas or metal and shall enclose the return side of the boom. The housing shall be attachable to and removable from the boom. Cutting or welding shall not be permitted.

3.15.3 Housing and supports over spirals. The housing shall be heavy canvas or metal and shall cover the feeder spirals or propellers. The center of the housing when made of canvas shall be split to allow a flap to drape over the side of a window. The housing shall be supported by a framework of steel angles and plate. Canvas shall be secured by bolts and nuts and metal shall be secured by rivets.

3.15.4 Outrigger type scraper runners. The scraper shoe runner connected to outrigger supports shall support the weight of the scraper when the loader is working on an unstable base. The outriggers shall be attached to the outer ends of the scraper by bolts.

3.15.5 Grade indicator. The grade indicator shall show the position of the scraper blade above or below grade in inch graduations. The indicator shall have a minimum range from plus 3 inches to minus 3 inches. The grade indicator shall be located on the side of the chassis in unobstructed view of the operator.

3.16 Brakes. Brakes shall be provided on the loader which will stop and hold the loader on a 20 percent grade with the loader facing upgrade and downgrade. With the engine running, the force required to apply each brake fully shall not exceed 50 pounds at the hand lever or foot pedal. Each brake shall lock the corresponding track with the above force applied. When the engine is shut down, the pedal force required to apply each brake, or both brakes when a single pedal is furnished, shall not exceed 150 pounds. Means for securing at least one brake in the applied position shall be provided for parking. When the parking brake is applied by a separate lever, the force required to actuate the lever, with the foot pedal depressed or lever pulled shall not exceed 90 pounds

3.17 Fuels, lubricants, and coolants. The loaders shall be assembled, run-in, and tested with the products of the type and grade specified in table I

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for the applicable ambient air temperature. The loader shall operate as specified herein when furnished with these products.

Table I. Service Products

Item	Application	Specification	Type & Grade
1.	Engine oils:		
	(a) Heavy duty	MIL-L-2104	10, 30, 50
	(b) Arctic	MIL-L-10295	One grade
	(c) High output diesel	MIL-L-45199	10, 30
2.	Fuel:		
	(a) Gasoline	MIL-G-3056	1 or 2
	(b) Diesel	VV-F-800	DF-1, DF-2, and DF-A
3.	Grease:		
	(a) Automotive	MIL-G-10924	One grade
	(b) High temperature	MIL-G-23827	One grade
4.	Hydraulic-system oil: (excl. brakes)		
	(a) Oil	MIL-L-2104	10
	(b) Oil (low temperature)	MIL-L-10295	One grade
5.	Lubricating oils:		
	(a) Universal	MIL-L-2105	75, 80, 90
	(b) Arctic	MIL-L-10324	One grade
	(c) Instrument	MIL-L-6085	One grade
6.	Antifreeze:		
	(a) Ethylene glycol	O-A-548	One grade
	(b) Arctic	MIL-A-11755	One grade

NOTE: At the option of the supplier, engines in production loaders may be serviced with high output diesel oils prior to delivery. The preproduction model testing shall be conducted with engine oil conforming to MIL-L-2104.

3.17.1 Lubrication fittings. Lubrication fittings shall be provided. Fittings shall be located in a protected position and shall be accessible to a grease gun with a flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operated quick-disconnect fasteners may be removed to provide accessibility.

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3.17.2 Pressure-release device. Where the use of pressure lubricating equipment will damage grease seals or other parts, a suitable pressure-release device shall be provided.

3.17.3 Delivery lubrication service. Unless otherwise specified herein, all components except those for which preservatives are specified in 5.1 shall be lubricated when delivered. Grease lubrication, including lubrication of sealed bearings, shall be with thermally stable greases conforming to MIL-G-10924 or MIL-G-23827. The loader shall be serviced with lubricants and grades of military oils (see table I) designated for use in the temperature range specified. Each loader shall be tagged in a conspicuous place to indicate the temperature range and grades of oils and greases used.

3.18 Fungus and moisture resistance. The electrical circuitry of all electrical systems, including all components and connections except as specified below, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173, except that 1 percent copper 8-quinolinolate (by weight) based on the non-volatile content of the varnish may be used in lieu of salicylanilide, the specified fungitoxicant:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1-mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.19 Identification marking. The loader shall be identified in accordance with MIL-STD-130. The marking shall be applied to the loader on identification plates conforming to MIL-P-514, type I, style 1, composition C of type I, grade A, class 1 material. Plates shall be furnished and mounted by the supplier. A shipping data plate shall be furnished and shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material, and in addition shall indicate the silhouette of the loader in transport position indicating the center of gravity. The plates shall be attached by screws, bolts, or rivets in conspicuous protected locations.

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3.19.1 Instruction plates. The loader shall be equipped with instruction plates including warnings, cautions, and diagrams describing any special or important procedures to be followed in assembling, operating, or servicing the loader. Instruction plates shall conform to MIL-P-519, type III, composition C, of type I, grade A, class 1 material. The plates shall be attached by screws, bolts, or rivets.

3.20 Treatment and painting. The portions of the loader (except the crankcase sealer) normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, color as specified. The application of a crankcase sealer as specified in MIL-T-704 on the interior surfaces need not be applied on the cast iron housings of transmission and torque converters provided the interior surfaces are void of casting defects such as blowholes, excessive porosity, cold shuts, hard spots, shrinks, cracks, or other injurious defects.

3.21 Stenciling. Registration and other marking shall be applied in accordance with MIL-STD-642. The gross weight and center-of-balance of the loader shall be stenciled on each side of the loader carrier.

3.22 Technical publications. Such technical publications as are specified shall be furnished.

3.23 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished.

3.24 Toolbox. The loader shall be provided with a toolbox made of mild steel with a nominal thickness of not less than 0.078 inch. The toolbox shall have an overlapping hinged lid which prevents the entry of rain and dirt, not less than two latches, and a hasp. The toolbox shall be of a size to accommodate the tools furnished (see 3.25) and shall be fastened to the loader in a protected location in the operator's compartment or be accessible to a man standing on the ground. The lid shall open at least 90 degrees from the closed position. The lid shall remain in the fully open position by the weight of the lid or by holding devices. Provisions shall be made to drain the toolbox in its mounted position.

3.25 Items to be furnished with loader. No shipment of loaders shall be made unless repair parts, maintenance tools, operating instructions, maintenance and repair manuals, parts lists, and accessories are included with the shipment or unless approval for shipment of the loaders without such items has been received from the contracting officer.

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3.26 Engine. Unless otherwise specified herein the engine, including all systems, components, accessories, and auxiliaries shall conform to MIL-E-11275, group II, or MIL-E-11276 as applicable. When specified, the engine shall be a standard commercial industrial type and shall include the systems, components, accessories, and auxiliaries specified in 3.26.1 through 3.26.9.

3.26.1 Power and speed rating. When a QPL engine is furnished, the power and speed rating of the engine shall be such that operation of the loader under any of the operational requirements and conditions specified herein will not require horsepower in excess of the intermittent horsepower rating of the engine for the applicable governed speed as established by the Qualified Products List, QPL-11275 or QPL-11276, as applicable. When a commercial-industrial engine is furnished, the power and speed rating of the engine shall be such that it shall provide sufficient power and torque to operate the loader under any of the operational requirements and conditions specified herein without exceeding the engine manufacturer's published, intermittent horsepower rating, or 90 percent of the engine manufacturer's published maximum horsepower rating (whichever is lower) at the applicable speeds.

3.26.2 Fuel tank. The fuel tank shall conform to MIL-T-46786 and be of sufficient capacity for not less than 8 hours of operation at the QPL continuous load consumption rate. A metal, bayonet-type dipstick, calibrated in increments of 1/4 tank capacity, shall be provided for the fuel tank in lieu of the mechanically actuated gage as specified in MIL-T-46786. The dipstick shall be coated or scored to clearly show the fuel level. A separate dustproof receptacle shall be provided in the top of the tank for the dipstick.

3.26.3 Governing system. The governing system shall conform to MIL-E-11275 or MIL-E-11276 as applicable, except that the speed at any load between no load and rated load shall be not less than the rated-load speed or shall not exceed the rated-load speed by more than 10 percent.

3.26.4 Cranking components. The engine-starting components shall include a waterproof starter button, solenoid-actuated heavy-duty electric starter motor, and flywheel drive mechanism. When tested as specified in 4.5.2.12, the system shall crank the engine at a speed of 150 rpm or at least 85 percent of the cranking rpm obtained during engine-qualification cold start in minus 25° F. ambient temperature, whichever is higher.

3.26.5 Alternator and regulator. A charging alternator (minimum rating, 35 amperes) and solid-state-type regulator shall be furnished.

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3.26.6 Batteries. Batteries shall conform to MS35000, size 6 TN.

3.26.7 Air cleaner. A dry-type air cleaner and a restriction indicator shall be provided. When the air cleaner is mounted horizontally, an automatic dump valve shall be provided on the primary separator to discharge the collected particles. The restriction indicator shall be located so that it is visible to the operator while in the normal operating position.

3.26.8 Exhaust system. A counter-balanced raincap shall be provided when the exhaust pipe is vertically mounted.

3.26.9 Neutral start. An attachment or control mechanism shall be provided within, or connected to, the shift linkage to prevent cranking of the engine except when the transmission is in neutral.

3.27 Electromagnetic interference. When specified, the electromagnetic interference emission characteristics of the loader shall conform to SAE J551.

3.28 Workmanship. The loader shall be free from defects such as incomplete welds, rust, cracks, and other defects that could impair its operation or serviceability.

3.28.1 Welders and welding.

3.28.1.1 Welder qualification. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by any of the following listed codes for the type of welding operation to be performed and that such qualification is effective by the particular code:

Standard Qualification Procedure of the American Welding Society.
Welding Qualification of the ASME Boiler and Pressure Vessel Code.

3.28.1.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings.

3.28.2 Steel fabrication. Steel used in the fabrication of the loader shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Corners shall be square and true, and all sharp edges and burrs shall be removed.

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Flame cutting may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground. Precautions shall be taken to avoid overheating, and heated metals shall be allowed to cool slowly. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.28.3 Bolted connections. Bolt holes shall be punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided and all bolts, nuts, and screws shall be tight.

3.28.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be full, concentric with the rivet hole, and in full contact with the surface of the member.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection. Prior to examination and test of the loader, the following shall be performed:

(a) When a QPL engine is furnished, test the engine to be installed in the preproduction model in accordance with MIL-STD-1400, Test Method Series 2,000, except as specified herein. This shall be

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accomplished on an engine dynamometer test stand by the engine manufacturer prior to installation in the loader and prior to delivery to the supplier. During Test Method 2200, check test and run-in, adjust the engine to deliver not more than the intermittent brake horsepower rating at the applicable speed as established by the engine qualification test. The fuel pump and governor as applicable shall be sealed to prevent adjustment during the preproduction tests. The reduced horsepower setting shall be retained throughout the preproduction model examination and tests. Any change made to this setting shall constitute failure of all tests. (After final acceptance of the preproduction model, the engine shall be re-adjusted to full-power setting.)

(b) Service the loader with the oils, greases, and fuel specified in table I and designated for use in the ambient temperature at which the tests will be run.

(c) Limber up the loader as prescribed by the supplier.

(d) Complete and submit to the contracting officer the Data Questionnaire (see 6.4).

4.3.1 Preproduction examination. Prior to testing, the loader shall be examined for the defects marked "X" in column 1 of table II. Presence of one or more defects shall be cause for rejection.

4.3.2 Preproduction tests. Upon successful completion of the examination specified in 4.3.1, the loader shall be subjected to the tests marked "X" in column 1 of table III. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Individual examination. After successful completion of the test specified in 4.4.2, each loader shall be examined for the defects marked "X" in column 2 of table II. Presence of one or more defects shall be cause for rejection.

4.4.2 Individual test. Each loader shall be tested as specified in column 2 of table III. Failure of the test shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. The examination shall be in accordance with table II.

4.5.2 Tests. Tests shall be conducted in accordance with table III under the conditions specified in 4.5.2.1.

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4.5.2.1 Test conditions.

(a) Test apparatus shall be accurate within the limits specified in SAE J872.
(b) Test methods referenced in the tests specified herein are those contained in the appendix.
(c) All working lights shall be turned on and left on during all performance tests.
(d) Unless otherwise specified herein, sequence of tests is optional with the supplier.
(e) Only that maintenance established by the supplier and submitted as a maintenance schedule prior to the tests shall be performed during the test.
(f) Evidence of breakage, malfunction, or permanent deformation occurring during any test shall constitute failure of that test, in addition to the failure criteria applicable to each test.

Table II. Examination Schedule

Pre-test	Qual. Assur.	Defect	Requirement Paragraph
1	2	3	4
X	X	101. Assembly not complete or incorrect.	Section 3
X	X	102. Lubrication devices not accessible or missing.	3.17 thru 3.17.3
X	-	103. Technical publications missing.	3.22
X	X	104. Replacement parts and tools missing or not processed.	3.23
		The following items not as specified:	
X	-	105. Boom assembly.	3.13 thru 3.13.3.2
X	-	106. Function of components and assemblies.	Section 3
X	-	107. Safety devices.	3.6
X	X	108. Material.	3.3
X	-	109. Electrical system	3.12 thru 3.12.2.5
X	-	110. Brake system.	3.16
X	-	111. Conveyor.	3.13.2
X	-	112. Discharge spout.	3.14
X	-	113. Operator's platform.	3.11

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Table II. Examination Schedule (cont'd)

1	2	3	4
X	-	114. Accessories.	3.15 thru 3.15.5
X	-	115. Controls.	3.11
X	-	116. Engine and accessories.	3.26 thru 3.26.9
X	X	117. Identification markings.	3.19 thru 3.19.3
X	X	118. Treatment and painting.	3.20
X	X	119. Stenciling.	3.21
X	X	120. Toolbox.	3.24
X	X	121. Workmanship.	3.28
-	X	122. Workmanship, materials, components or adjustments different from accepted preproduction model.	Section 3

Table III. Test Schedule

Pre-pro.	Qual. Assur.	Test	T.M. No.	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
X	-	Stockpile loading and capacity.	1	4.5.2.2	3.1
X	-	Windrow loading, variable loading speed.	2	4.5.2.3	3.1
X	X	Travel speed.	3	4.5.2.4	3.8.1 & 3.8.2
X	-	Service brakes, parking brake.	4	4.5.2.5	3.16
X	-	Loading speed.	5	4.5.2.6	3.8.2
X	-	Electromagnetic interference.	-	4.5.2.7	3.27
X	-	Rain - wind.	6	4.5.2.8	3.5.2
X	-	Discharge spout.	7	4.5.2.9	3.14
X	-	Conveyor speed.	8	4.5.2.10	3.13.2
X	-	Turning radius.	9	4.5.2.11	3.9
X	-	Environmental.	10&11	4.5.2.12	3.5.1
X	-	Boom action.	12	4.5.2.13	3.13.1
-	X	Operational.	13	4.5.2.14	Section 3

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4.5.2.2 Stockpile loading and capacity. Test the loader in accordance with test method No. 1. Nonconformance to 3.1 shall constitute failure of this test.

4.5.2.3 Windrow loading, variable loading speed. Test the loader in accordance with test method No. 2. Nonconformance with 3.1 shall constitute failure of this test.

4.5.2.4 Travel speed. Test the loader in accordance with test method No. 3. Nonconformance with 3.8.1 and 3.8.2 shall constitute failure of this test.

4.5.2.5 Brakes. Test the loader in accordance with test method No. 4. Nonconformance to 3.16 shall constitute failure of this test.

4.5.2.6 Loading speed. Test the loader in accordance with test method No. 5. Nonconformance with 3.8.2 shall constitute failure of this test.

4.5.2.7 Electromagnetic interference. Test the loader in accordance with SAE J551. The supplier shall furnish to the contracting officer a test report. Disapproval of the report shall constitute failure of this test.

4.5.2.8 Rain-wind. Test the loader in accordance with test method No. 6. Nonconformance with 3.5.2 shall constitute failure of this test.

4.5.2.9 Discharge spout. Test the loader in accordance with test method No. 7. Nonconformance with 3.14 shall constitute failure of this test.

4.5.2.10 Conveyor speed. Test the loader in accordance with test method No. 8. Nonconformance with 3.13.2 shall constitute failure of this test.

4.5.2.11 Turning radius. Test the loader in accordance with test method No. 9. Nonconformance with 3.9 shall constitute failure of this test.

4.5.2.12 Environmental. Test the loader in accordance with test methods No. 10 and 11. Nonconformance with 3.5.1 shall constitute failure of this test.

4.5.2.13 Boom action. Test the loader in accordance with test method No. 12. Nonconformance with 3.13.1 shall constitute failure of this test.

4.5.2.14 Operational. Test the loader in accordance with test method No. 13. Evidence of any damage, malfunction, or incorrect adjustment which shall preclude continued normal operation shall constitute failure of this test.

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4.6 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be examined and tested to determine compliance with MIL-T-3351.

5. PREPARATION FOR DELIVERY.

5.1 Preservation, packaging, packing, and marking. Each complete loader with repair parts, tools, publications, and attachments shall be preserved, packaged, packed, and marked as specified for crawler-mounted tractors and their attachments in accordance with MIL-T-3351, type II and III classification. The level of preservation and packaging shall be level A, B, or C and the level of packing shall be level A, B, or C, as specified.

6. NOTES

6.1 Intended use. The loader is intended for use primarily for loading loose material such as gravel, sand, and cold-mix asphalt from a stockpile or windrows or for loading directly from a finished surface of subgrade.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class of loader required (see 1.2).
(c) Time frame required for submission of the preproduction model (see 3.2).
(d) When crawlers shall be provided with bolt holes for grouser attachment (see 3.9).
(e) When accessories are required and which accessories are required (see 3.15 through 3.15.5).
(f) Temperature range required for delivery lubrication service (see 3.17.3).
(g) Color of paint required (see 3.20).
(h) Technical publications required (see 3.22).
(i) Repair parts and maintenance tools required (see 3.23).
(j) When a commercial engine is required (see 3.26).
(k) When electromagnetic interference reduction is required (see 3.27).
(l) Level of preservation and packaging and level of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production loaders from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish loaders conforming to this specification.

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6.4 Data questionnaire. The contracting officer should require the supplier to complete and submit the questionnaire, part 2 of the appendix of this specification, prior to preproduction model testing as evidence of compliance with the requirements.

Custodians:

Army - ME
Navy - YD

Review activities:

Army - SM

User activities:

Army - CE
Navy - MC

Preparing activity:

Army - ME

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APPENDIX

PART I

TEST METHODS

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TEST METHOD NO. 1

STOCKPILE LOADING AND CAPACITY

1. TEST COURSE:

The test course shall be a level hardstand or storage area with sufficient area for three separate stockpiles.

2. TEST MATERIAL:

(a) Building sand or washed river sand.
(b) 1-inch minus gravel.
(c) 2-1/2-inch minus gravel.

3. TEST APPARATUS:

(a) Two dump trucks of different height.
(b) 1 cubic foot measure.
(c) Stopwatch.
(d) Truck scale.
(e) Platform scale.

4. TEST PROCEDURE:

(a) Determine loose density per cubic yard of the sand and gravel by weight; the 1 cubic foot measure filled with each, separately, struck, not compacted.
(b) Spot the trucks side by side and parallel to the longitudinal axis of the loader. Load each truck with sand by pivoting the swivel spout and without changing the direction of the truck or the loader.

(1) Weigh empty trucks and record weight.
(2) Record loading time for each load.
(3) Weigh each loaded truck, dump load, and reload. Record weight.
(4) Weigh empty trucks. Record weight.

(c) Using the density of the sand, total net weight of the 4 truck loads, and total elapsed time required to load, determine the loading rate in cubic yards per minute.
(d) Repeat 4b and 4c using 1-inch minus gravel.
(e) Repeat 4b and 4c using 2-1/2-inch minus gravel.

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TEST METHOD NO. 2

WINDROW LOADING AND ACCESSORIES

1. TEST COURSE:

The test area shall be of sufficient size in which the loader can operate in conjunction with the windrow as described below.

2. TEST MATERIAL:

A windrow of 2-1/2-inch minus crushed stone which shall be not less than 300 feet long and shall vary in size from 10 square feet cross-sectional area to 2 square feet cross-sectional area.

3. TEST EQUIPMENT:

(a) Tape, measuring.
(b) Stopwatch.
(c) Stakes.

4. TEST PROCEDURE:

(a) Operate the loader by loading the windrow in each crowd speed for a distance of not less than 50 feet. Measure the distance traversed and the time elapsed in each crowd speed between the stakes.
(b) Determine the speed in each crowd speed in fpm.
(c) While loading the windrow, operate the discharge spout in the farthest right position, farthest left position, and in line with the longitudinal axis. Operate in each position for not less than 100 feet.
(d) Inspect the operation of the accessories, when furnished during the test.

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TEST METHOD NO. 3

TRAVEL SPEED

1. TEST COURSE:

The test course shall be a dry level, improved surface of sufficient length to permit the loader to attain maximum speed prior to entering the measured distance. The measured portion of the course shall be not less than 100 feet long.

2. TEST APPARATUS:

(a) Tape, measuring.
(b) Stopwatch.

3. TEST PROCEDURE:

(a) Operate the loader at not more than full-load governed speed.
(b) Drive the loader a sufficient distance to attain maximum speed prior to entering the measured test course.
(c) Loader speed shall be the average of 2 runs, 1 in each direction.
(d) Conduct the following tests:

(1) Speed - high gear forward.
(2) Speed - low gear forward.
(3) Speed - reverse gear.

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TEST METHOD NO. 4

BRAKES

1. TEST COURSE:

Paved or compacted slope with a grade of not less than 20 percent. Length of the slope shall be greater than the overall length of the loader.

2. TEST APPARATUS:

Spring scale or other device to measure force applied to foot pedals or levers.

3. TEST PROCEDURE:

(a) Drive the loader forward halfway up the slope. Stop and hold the loader with the service brakes, set the parking brake, release the service brake and dismount.
(b) Record pressure required to apply the brakes.
(c) Remount and repeat the procedures except with the loader facing down slope.

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TEST METHOD NO. 5

LOADING SPEED

1. TEST COURSE:

The test course shall be the same as test method No. 3 except the measured portion of the course shall be not less than 50 feet long.

2. TEST APPARATUS:

(a) Tape, measuring.
(b) Stopwatch.

3. TEST PROCEDURE:

(a) Operate the loader at not more than full-load governed speed.
(b) Drive the loader a sufficient distance to attain maximum speed prior to entering the measured test course.
(c) Loader speed shall be the average of 2 runs, 1 in each direction.
(d) Repeat the above for each gear.

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TEST METHOD NO. 6

RAIN-WIND

1. TEST APPARATUS:

   (a) Timing device.
   (b) Metering device.
   (c) Spray fixture.
   (d) Wind generator and wind velocity gage.
   (e) A means to trap and collect water which may enter the exhaust system leading to the exhaust manifold.

2. TEST PROCEDURE:

   (a) Operating condition.

      (1) Park the loader on a horizontal surface and expose to simulated rain falling toward the loader at an angle of approximately 45 degrees to the horizontal. The volume of rain impinging on the horizontal surface shall be not less than 1/4 gallon per minute per square foot of wetted, horizontal area (24 inches of rain per hour). Simulated rain shall originate from a source which produces droplets of uniform distribution and velocity similar to the natural rain at the point of contact with the loader. The rain shall then be driven horizontally onto and against the center of mass of the loader by a generated wind of not less than 30 miles per hour.

   (b) Storage condition.

      (1) Position the loader to expose the exhaust opening to the most critical direction with respect to the wind for water infiltration. Shut down engine.
      (2) Insert item 1(e) in the exhaust system at a location where it will trap and collect water entering into the exhaust system.
      (3) Expose the loader in this position to the rain as described above for a period of 5 minutes.
      (4) Stop the wind and rain.
      (5) Examine the water-collecting means for presence of water.

   (c) Operating.

      (1) Expose the top and left side to the rain as described above for a period of 5 minutes with the engine shut down.

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(2) With water flowing, start engine within 2 minutes (without removing accumulated water) and allow engine to idle for 2 minutes.
(3) Repeat (1) and (2) with the top and front exposed.
(4) Repeat (1) and (2) with the top and right side exposed.
(5) Repeat (1) and (2) with the top and rear exposed.

(d) At the conclusion of the test, examine the loader for the following and record observations:

(1) Water in the air-intake filter.
(2) Water leakage into the engine crankcase sump, exhaust system, and hydraulic reservoir.
(3) Performance of the electrical components, gages, and instruments.

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TEST METHOD NO. 7

DISCHARGE SPOUT SWING

1. TEST COURSE:

The test course shall be a dry level surface of sufficient area for the loader and for personnel to move around the loader.

2. TEST APPARATUS:

(a) Plumb bob.
(b) Protractor.
(c) String.

3. TEST PROCEDURE:

(a) Center the discharge spout over the longitudinal axis of the loader.
(b) Swing the spout to maximum right swing. Measure and record the angle included.
(c) Center the discharge spout over the longitudinal axis of the loader.
(d) Swing the spout maximum left swing. Measure the angle and record.

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TEST METHOD NO. 8

CONVEYOR SPEED

1. TEST COURSE:

The test area shall be essentially level with sufficient area for either a windrow or stockpile.

2. TEST MATERIAL:

The material to be loaded shall be friable (noncohesive), e.g. bank run gravel, crushed rock, or building sand, deposited in a stockpile or windrow.

3. TEST EQUIPMENT:

(a) Tape, measuring.
(b) Stopwatch.
(c) Paint.

4. TEST PROCEDURE:

(a) Paint a mark on the chain.
(b) Measure the length of the chain.
(c) Operate the loader under maximum capacity on either the stockpile or windrow and count the number of cycles the chain makes in one minute, using the point mark as a reference point.
(d) Determine the speed in rpm of the conveyor chain by multiplying the cycles per minute times the chain length.

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TEST METHOD NO. 9

TURNING RADIUS

1. TEST COURSE:

The test area shall be an essentially level unpaved surface of sufficient size to permit the loader to turn.

2. TEST APPARATUS:

(a) Tape, measuring.
(b) Stick, pointed.

3. TEST PROCEDURE:

(a) Determine the centerline of the right track and mark its extension upon the ground.
(b) Turn about the left track in a forward direction for 180 degrees. Make another mark in the same manner and measure the distance between the two parallel lines.
(c) To determine the turning radius, half the measured distance.
(d) Repeat the above procedure turning about the right track.

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TEST METHOD NO. 10

LOW TEMPERATURE

1. TEST APPARATUS:

(a) Cold chamber.
(b) Arctic clothing.
(c) Thermocouples.
(d) Battery hydrometer.
(e) Wattmeter or ammeter.
(f) Recording instruments (temperature reading, electrical).
(g) Voltmeter.

2. TEST PROCEDURE:

(a) Subject the loader with accessories to an ambient temperature of minus 25° F. or lower until all parts, lube oil, coolant, and hydraulic oil have stabilized at this temperature. Determine the temperature by taking temperature readings from thermocouples located as follows: (a) In the two center cells of each battery between the plates and at middepth of the electrolyte; (b) in the engine lubricating oil gallery; (c) in the engine coolant in the engine-cooling-jacket discharge pipe; (d) in the chamber at a horizontal angle of 45 degrees from the sides and ends of the loader and not less than 3 feet from each corner of the loader at engine-crankshaft level; (e) in the operator's compartment 6 inches above the floor boards and adjacent to the foot pedals. Maintain an ambient temperature of minus 25° F. or lower throughout the entire test. Substitute or supplementary batteries or other electrical cranking energy shall not be used. Glow plugs or other accessories working off the cranking batteries and auxiliary fluid priming systems which are permanently mounted on the loader and normally used in starting the engine may be used. One operator wearing arctic gloves shall perform all priming and starting operations.
(b) Check the specific gravity of each battery.
(c) Crank the engine using batteries that were cold soaked and stabilized at minus 25° F. or lower.
(d) Start the engine. Measure and record the rpm of the engine during the cranking period. The engine should run on the fuel in not more than 5 minutes after beginning of first cranking (excluding starting engine). Smooth engine running without unnatural or continued control manipulation and without use of primers and priming fluids should be obtained within 15 minutes after starting.

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(e) Record temperatures prior to engine starting, immediately after engine starting, and at 10-minute intervals during engine warmup and cycling, including but not limited to the following temperatures: (1) Battery, (2) hydraulic tank, (3) engine oil, (4) engine coolant, (5) chamber air, and (6) operator's compartment 6 inches above the floor level between or adjacent to the foot pedals.

(f) After the engine has started, run the engine for 15 minutes. Then operate the engine as follows: (1) 10 minutes idling, (2) 10 minutes half throttle, and (3) 5 minutes full throttle. Perform the 25-minute cycle a total of three times.

(g) During engine cycling and while wearing arctic gloves and boots, perform each of the following: (1) Engage and disengage power train to the tracks; (2) operate through two complete movement cycles all clutches, brakes, shafts, drums, and cylinders; and (3) mount to and dismount from the operator's position on the loader.

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TEST METHOD NO. 11

HIGH TEMPERATURE

1. TEST APPARATUS:

(a) Test chamber.
(b) Thermocouples.
(c) Recording potentiometer(s).
(d) Thermometers (wet and dry).
(e) Dynamometer with all necessary instrumentation.
(f) Tachometer.
(g) Barometer.

2. TEST PROCEDURE:

(a) Subject the loader with accessories to an ambient temperature of plus 120° F. until the engine coolant, fuel in the fuel tank, engine lubricating oil gallery, and batteries have stabilized at that temperature. Determine the temperature by taking temperature readings from thermocouples located as follows: (1) One in the engine-cooling-jacket discharge and another in the coolant inlet to the engine; (2) in the fuel tank; (3) in the two center cells of each battery between the plates and at middepth of the electrolyte; (4) in the chamber at a horizontal angle of 45 degrees from the sides and ends of the loader and not less than 3 feet from each corner of the loader at engine-crankshaft level; (5) in the engine lubricating oil gallery. Disconnect the drive train at a convenient joint and connect the dynamometer to the drive train. Calculate the dynamometer loads required to develop the engine loads required in 3.26.1.
(b) Start the engine.
(c) Apply a load equal to 90 percent of the intermittent brake horsepower at governed speed for 1 hour. At the end of the 1-hour period, change the load and cycle as follows for 10 complete cycles:

(1) Maximum rated load (as determined above) - 3 minutes.
(2) One-half maximum rated load - 2 minutes.
(3) Zero load - 1 minute.

(d) During the 2-hour operation specified in (b), record at 10-minute intervals temperature readings of the thermocouples specified in 2(a). Also record at 10-minute intervals the following:

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(1) Revolutions per minute.
(2) Dynamometer torque. (Take readings (1) and (2) simultaneously).
(3) Barometer.
(4) Wet- and dry-bulb air temperatures.

(e) Operate all controls and switches other than the starting switch.

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TEST METHOD NO. 12

BOOM ACTION

1. TEST COURSE:

The test course shall be a dry, level surface of sufficient area to contain the loader.

2. TEST PROCEDURE:

Fully raise and lower the loader boom not less than six times.

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TEST METHOD NO. 13

OPERATIONAL

1. TEST COURSE:

Paved or improved level area of sufficient size to accommodate operation of the loader as described in the test procedure.

2. TEST PROCEDURE:

(a) Operate the loader continuously for not less than 30 minutes, not loading material.

(1) Manipulate each control on the loader not less than 10 times. Check for adjustment and ease of operation.
(2) Check each gage, instrument, and indicator for proper functioning.
(3) Note and record all indications of maladjustment, faulty operation, and malfunction.

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APPENDIX

PART II

DATA QUESTIONNAIRE

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QUESTIONNAIRE FOR PHYSICAL AND FUNCTIONAL DATA

PART I - DESCRIPTION

The loader submitted is described as follows:

Supplier

Supplier's model designation

Serial number range

A. ENGINE.

Make

Model

Series

Type

Cycles

No. cylinders

Flywheel horsepower ratings (at RPMs):

	Commercial	QPL	As Installed
Maximum	at	at	at
Intermittent	at	at	at
Continuous	at	at	at

Bore

Stroke

Compression ratio

Air induction system

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Suppliers shall furnish certified torque and net horsepower output curves for continuous duty.

Suppliers shall furnish certified converter performance curves for engine-converter output.

B. ENGINE ACCESSORIES.

1. Air cleaner

(a) Make

(b) Model

(c) Type

2. Oil filter

(a) Type

(b) Make

(c) Housing No.

(d) Element No.

3. Fuel filter

(a) Type

(b) Make

(c) Element No. ; FSN

4. Temperature gage

(a) Type

(b) Make

(c) Range

(d) Location degrees F. to degrees F.

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5. Fuel pump

   (a) Type

   (b) Make

   (c) Capacity

6. Hourmeter

   (a) Type

   (b) Actuated by

   (c) Make

   (d) Model

7. Governor

   (a) Type

   (b) Make

   (c) Model

   (d) Setting for engine rpm.

8. Muffler

   (a) Type

   (b) Make

9. Starting system, electric

   (a) Type

   (b) Voltage

   (c) Make

   (d) Model

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10. Lubricating oil pressure gage

(a) Type

(b) Range to psi.

11. Hand throttle (if applicable)

(a) Type

(b) Location

C. COOLING SYSTEM.

1. General

(a) Type

(b) Overall capacity gal.

(c) Pressurization psi.

2. Radiator

(a) Type

(b) Capacity gal.

(c) Make

(d) Protective guard

3. Fan

(a) Type

(b) Diameter in.

(c) No. blades

(d) Blade spacing

4. Water pump

(a) Type

(b) Make and model

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(c) Drive

(d) Rating gpm at rpm

(e) Speed as installed rpm; gpm

5. Thermostat

(a) Type

(b) Make and model

(c) Temperature rating: Full open degrees F.;

Closed degrees F.

D. POWER TRAIN.

1. Transmission

(a) Type

(b) Make

(c) Speed range

(d) Gear ratios:

Forward 1. 2.

Reverse 1.

2. Torque converter (if applicable)

(a) Make

(b) Model

(c) Torque input Rated Actual

(d) Rpm

(e) Hp input Rated Actual

(f) Torque multiplication factor

(g) Stall torque ratio

(h) Heat Exchanger

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3. Master clutch (if applicable)

(a) Make

(b) Model

(c) Type

(d) Size

(e) Facing area

(f) Torque capacity lb. ft.

(g) Pilot bearing

(h) Throw out bearing

(i) Friction surface pressure psi.

4. Steering clutches

(a) Make

(b) Model

(c) Type

(d) Size

(e) Facing area

(f) Torque capacity lb. ft.

(g) Pilot bearing

(h) Throw out bearing

(i) Friction surface pressure psi.

E. FINAL DRIVE.

(a) Type drive

(b) Final drive gear ratio

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F. BRAKES.

1. Service brakes

(a) Type

(b) Make

(c) Model

(d) No. brake drums Diameter

Lining width

Brake chamber volume cu. in.

(e) Location

(f) Air compressor: Type Capacity cfm

Type drive Location

(g) Reservoir size cu. in.

(h) Pressure gage location

Operating pressure psi

2. Parking brake

(a) Type

(b) Make

(c) Model

(d) Actuating

(e) Control: Type Located

G. CONTROLS.

1. Operating

(a) Type

(b) Overload protection

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2. Steering
   (a) Type
   (b) Make
   (c) Model
   (d) Location
3. Hydraulic system
   (a) Make
   (b) Tank capacity gal.
   (c) Filler location
   (d) Oil tank breather location
4. High pressure lines (hydraulic)
   (a) Type
   (b) Protection
5. Pressure relief valve
   (a) Type
   (b) Make
   (c) Piston size
   (d) Pressure setting
6. Pump
   (a) Type
   (b) Make
   (c) Model
   (d) Drive
   (e) Drive ratio, engine to pump

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(f) Capacity: Rated Actual

(1) Gpm

(2) At rpm

(3) Psi

7. Filter

(a) Make

(b) Model

(c) Element

H. OPERATOR'S AREA.

1. Type

2. Dimensions

3. Material

I. ELECTRICAL EQUIPMENT.

1. Battery

(a) No.

(b) Size

(c) Volts each

(d) Make

(e) Rating amp hr at 20-hr rate

(f) No. plates

(g) Mounting

(h) Location

2. Alternator

(a) Type

(b) Volts

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(c) Rating amperes rpm

(d) Make

(e) Model

(f) Serial No.

(g) Drive

(h) Control: Type

Make

Model

(i) Reverse polarity protection

3. Horn (if applicable)

(a) Type

(b) Make

(c) Button location

4. Starter

(a) Type

(b) Make

(c) Model

(d) Serial No.

5. Electrical wiring

(a) Type

(b) Protection

J. GAGES AND INSTRUMENTS.

1. Engine oil pressure

2. Engine temperature gage

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3. Engine hourmeter

4. Ammeter

(a) Make

(b) Range

(c) Location

5. Torque converter temperature gage (if applicable)

(a) Make

(b) Model

(c) Range

(d) Location

6. Air pressure gage (when applicable)

(a) Make

(b) Range

(c) Location

K. TOOLBOX.

(a) Type

(b) No.

(c) Dimensions

(d) Location(s)

(e) Cover fastening

L. FUEL TANK.

(a) Capacity gal.

(b) Location

(c) Drain plug location

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PART II. PHYSICAL AND FUNCTIONAL DIMENSIONS

A. PHYSICAL DIMENSIONS

Suppliers shall furnish profile working drawing of end item, side and end views.

1. Shipping weight and std. equipment, dry lb.
2. Operating weights with fuel, coolant, oil, grease, hydraulic fluid, etc.

   Right side lb.

   Left side lb.

   Total lb.

3. Track mechanism
   (a) Vertical oscillation inches
   (b) No. track rollers inches
   (c) No. track carrier rollers
   (d) Shoes
      (1) Type
      (2) Width
      (3) Number each track
      (4) Grouser height inches
   (e) Gage inches
   (f) Ground contact in^2
4. Turning clearance diameter.
   (a) Right turn, feet
   (b) Left turn, feet

Turning diameter is defined as the diameter of the smallest circle described by the outermost part of the loader in making a 180-degree turn.

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5. (a) Overall length, loading position in.

(b) Overall width in.

(c) Width of feeding system overall in.

(d) Overall height, operating, in.

(e) Overall height, reducible in.

PART III. Supplier shall furnish detailed list of all items which vary from his commercial loader.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION **MIL-L-403D-Loaders, Bucket-Type**

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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